

audio-classification

May 3, 2025

```
[6]: !pip install librosa
!pip install tensorflow
!pip install matplotlib
!pip install pandas
!pip install tqdm
!pip install scikit-learn
!pip install seaborn
!pip install scikit-image
import librosa
import IPython.display as ipd #to visualize the sound data
import librosa.display
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

Requirement already satisfied: librosa in ./env/lib/python3.10/site-packages (0.11.0)

Requirement already satisfied: numpy>=1.22.3 in ./env/lib/python3.10/site-packages (from librosa) (2.1.3)

Requirement already satisfied: joblib>=1.0 in ./env/lib/python3.10/site-packages (from librosa) (1.4.2)

Requirement already satisfied: soundfile>=0.12.1 in ./env/lib/python3.10/site-packages (from librosa) (0.13.1)

Requirement already satisfied: numba>=0.51.0 in ./env/lib/python3.10/site-packages (from librosa) (0.61.2)

Requirement already satisfied: lazy_loader>=0.1 in ./env/lib/python3.10/site-packages (from librosa) (0.4)

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Requirement already satisfied: pooch>=1.1 in ./env/lib/python3.10/site-packages (from librosa) (1.8.2)

Requirement already satisfied: soxr>=0.3.2 in ./env/lib/python3.10/site-packages (from librosa) (0.5.0.post1)

Requirement already satisfied: scipy>=1.6.0 in ./env/lib/python3.10/site-packages (from librosa) (1.15.2)

Requirement already satisfied: audioread>=2.1.9 in ./env/lib/python3.10/site-packages (from librosa) (3.0.1)

Requirement already satisfied: scikit-learn>=1.1.0 in ./env/lib/python3.10/site-packages (from librosa) (1.6.1)

Requirement already satisfied: decorator>=4.3.0 in ./env/lib/python3.10/site-packages (from librosa) (5.2.1)

Requirement already satisfied: packaging in ./env/lib/python3.10/site-packages (from lazy_loader>=0.1->librosa) (25.0)

Requirement already satisfied: llvmlite<0.45,>=0.44.0dev0 in ./env/lib/python3.10/site-packages (from numba>=0.51.0->librosa) (0.44.0)

Requirement already satisfied: requests>=2.19.0 in ./env/lib/python3.10/site-packages (from pooch>=1.1->librosa) (2.32.3)

Requirement already satisfied: platformdirs>=2.5.0 in ./env/lib/python3.10/site-packages (from pooch>=1.1->librosa) (4.3.7)

Requirement already satisfied: threadpoolctl>=3.1.0 in ./env/lib/python3.10/site-packages (from scikit-learn>=1.1.0->librosa) (3.6.0)

Requirement already satisfied: cffi>=1.0 in ./env/lib/python3.10/site-packages (from soundfile>=0.12.1->librosa) (1.17.1)

Requirement already satisfied: pycparser in ./env/lib/python3.10/site-packages (from cffi>=1.0->soundfile>=0.12.1->librosa) (2.22)

Requirement already satisfied: certifi>=2017.4.17 in ./env/lib/python3.10/site-packages (from requests>=2.19.0->pooch>=1.1->librosa) (2025.4.26)

Requirement already satisfied: charset-normalizer<4,>=2 in ./env/lib/python3.10/site-packages (from requests>=2.19.0->pooch>=1.1->librosa) (3.4.2)

Requirement already satisfied: urllib3<3,>=1.21.1 in ./env/lib/python3.10/site-packages (from requests>=2.19.0->pooch>=1.1->librosa) (2.4.0)

Requirement already satisfied: idna<4,>=2.5 in ./env/lib/python3.10/site-packages (from requests>=2.19.0->pooch>=1.1->librosa) (3.10)

Requirement already satisfied: tensorflow in ./env/lib/python3.10/site-packages (2.19.0)

Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in ./env/lib/python3.10/site-packages (from tensorflow) (0.6.0)

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Requirement already satisfied: packaging in ./env/lib/python3.10/site-packages (from tensorflow) (25.0)

Requirement already satisfied: six>=1.12.0 in ./env/lib/python3.10/site-packages (from tensorflow) (1.17.0)

Requirement already satisfied: astunparse>=1.6.0 in ./env/lib/python3.10/site-packages (from tensorflow) (1.6.3)

Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0dev,>=3.20.3 in ./env/lib/python3.10/site-packages (from tensorflow) (5.29.4)

Requirement already satisfied: numpy<2.2.0,>=1.26.0 in ./env/lib/python3.10/site-packages (from tensorflow) (2.1.3)

Requirement already satisfied: h5py>=3.11.0 in ./env/lib/python3.10/site-packages (from tensorflow) (3.13.0)

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Requirement already satisfied: requests<3,>=2.21.0 in ./env/lib/python3.10/site-packages (from tensorflow) (2.32.3)

Requirement already satisfied: grpcio<2.0,>=1.24.3 in ./env/lib/python3.10/site-packages (from tensorflow) (1.71.0)

Requirement already satisfied: flatbuffers>=24.3.25 in ./env/lib/python3.10/site-packages (from tensorflow) (25.2.10)

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Requirement already satisfied: absl-py>=1.0.0 in ./env/lib/python3.10/site-packages (from tensorflow) (2.2.2)

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Requirement already satisfied: libclang>=13.0.0 in ./env/lib/python3.10/site-packages (from tensorflow) (18.1.1)

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Requirement already satisfied: namex in ./env/lib/python3.10/site-packages (from keras>=3.5.0->tensorflow) (0.0.9)

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Requirement already satisfied: certifi>=2017.4.17 in ./env/lib/python3.10/site-packages (from requests<3,>=2.21.0->tensorflow) (2025.4.26)

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Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in ./env/lib/python3.10/site-packages (from tensorboard~=2.19.0->tensorflow) (0.7.2)

Requirement already satisfied: markdown>=2.6.8 in ./env/lib/python3.10/site-packages (from tensorboard~=2.19.0->tensorflow) (3.8)

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Requirement already satisfied: MarkupSafe>=2.1.1 in ./env/lib/python3.10/site-packages (from werkzeug>=1.0.1->tensorboard~=2.19.0->tensorflow) (3.0.2)

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Requirement already satisfied: matplotlib in ./env/lib/python3.10/site-packages (3.10.1)

Requirement already satisfied: kiwisolver>=1.3.1 in ./env/lib/python3.10/site-packages (from matplotlib) (1.4.8)

Requirement already satisfied: pillow>=8 in ./env/lib/python3.10/site-packages (from matplotlib) (11.2.1)

Requirement already satisfied: fonttools>=4.22.0 in ./env/lib/python3.10/site-packages (from matplotlib) (4.57.0)

Requirement already satisfied: pyparsing>=2.3.1 in ./env/lib/python3.10/site-packages (from matplotlib) (3.2.3)

Requirement already satisfied: packaging>=20.0 in ./env/lib/python3.10/site-packages (from matplotlib) (25.0)

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Requirement already satisfied: contourpy>=1.0.1 in ./env/lib/python3.10/site-packages (from matplotlib) (1.3.2)

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Requirement already satisfied: six>=1.5 in ./env/lib/python3.10/site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

Requirement already satisfied: pandas in ./env/lib/python3.10/site-packages (2.2.3)

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Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in ./env/lib/python3.10/site-packages (from seaborn) (3.10.1)

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Requirement already satisfied: python-dateutil>=2.7 in ./env/lib/python3.10/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)

Requirement already satisfied: packaging>=20.0 in ./env/lib/python3.10/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (25.0)

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Requirement already satisfied: pyparsing>=2.3.1 in ./env/lib/python3.10/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.3)

Requirement already satisfied: pytz>=2020.1 in ./env/lib/python3.10/site-packages (from pandas>=1.2->seaborn) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in ./env/lib/python3.10/site-packages (from pandas>=1.2->seaborn) (2025.2)

Requirement already satisfied: six>=1.5 in ./env/lib/python3.10/site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)

Requirement already satisfied: scikit-image in ./env/lib/python3.10/site-packages (0.25.2)

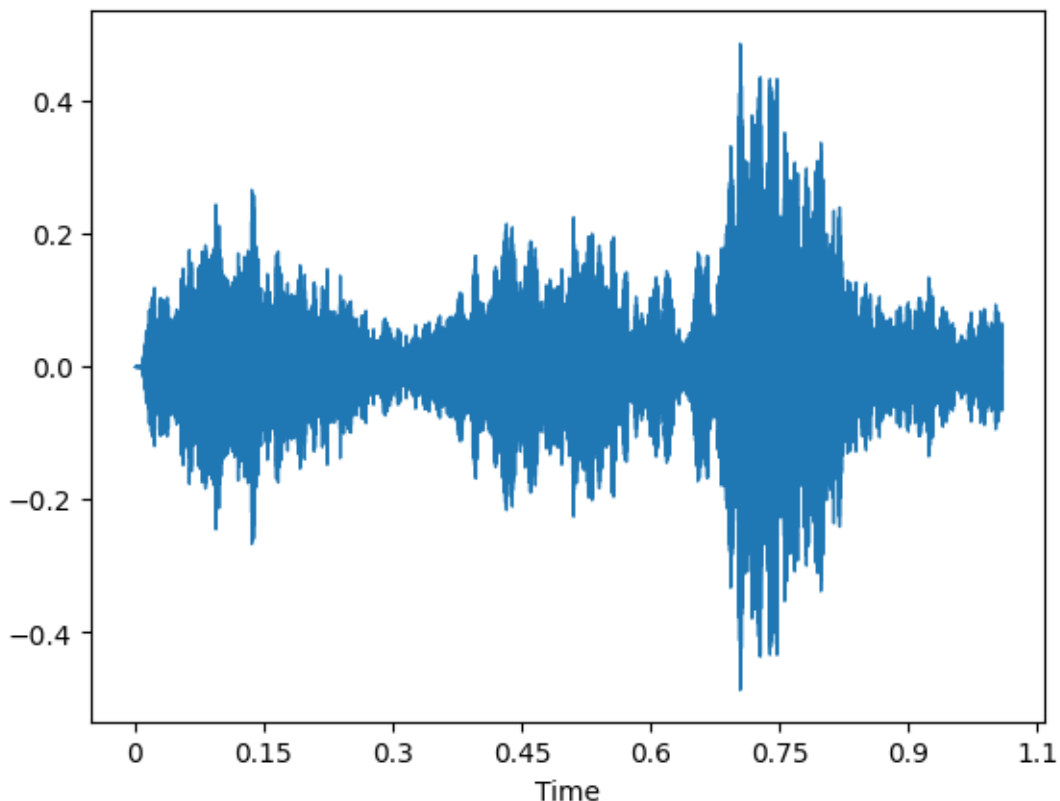
Requirement already satisfied: pillow>=10.1 in ./env/lib/python3.10/site-packages (from scikit-image) (11.2.1)

Requirement already satisfied: lazy-loader>=0.4 in ./env/lib/python3.10/site-packages (from scikit-image) (0.4)

Requirement already satisfied: scipy>=1.11.4 in ./env/lib/python3.10/site-packages (from scikit-image) (1.15.2)
Requirement already satisfied: networkx>=3.0 in ./env/lib/python3.10/site-packages (from scikit-image) (3.4.2)
Requirement already satisfied: tifffile>=2022.8.12 in ./env/lib/python3.10/site-packages (from scikit-image) (2025.3.30)
Requirement already satisfied: imageio!=2.35.0,>=2.33 in ./env/lib/python3.10/site-packages (from scikit-image) (2.37.0)
Requirement already satisfied: packaging>=21 in ./env/lib/python3.10/site-packages (from scikit-image) (25.0)
Requirement already satisfied: numpy>=1.24 in ./env/lib/python3.10/site-packages (from scikit-image) (2.1.3)

```
[8]: filename='./audio/fold2/4201-3-3-0.wav' # select a random oneplt.  
     ↪figure(figsize=(14,5))  
     data,sample_rate=librosa.load(filename)  
     librosa.display.waveshow(data,sr=sample_rate)  
     ipd.Audio(filename)
```

[8]: <IPython.lib.display.Audio object>



```
[9]: metadata=pd.read_csv('audio/UrbanSound8K.csv')
      metadata.head(10)
```

```
[9]:
```

	slice_file_name	fsID	start	end	salience	fold	classID	\
0	100032-3-0-0.wav	100032	0.000000	0.317551	1	5	3	
1	100263-2-0-117.wav	100263	58.500000	62.500000	1	5	2	
2	100263-2-0-121.wav	100263	60.500000	64.500000	1	5	2	
3	100263-2-0-126.wav	100263	63.000000	67.000000	1	5	2	
4	100263-2-0-137.wav	100263	68.500000	72.500000	1	5	2	
5	100263-2-0-143.wav	100263	71.500000	75.500000	1	5	2	
6	100263-2-0-161.wav	100263	80.500000	84.500000	1	5	2	
7	100263-2-0-3.wav	100263	1.500000	5.500000	1	5	2	
8	100263-2-0-36.wav	100263	18.000000	22.000000	1	5	2	
9	100648-1-0-0.wav	100648	4.823402	5.471927	2	10	1	


```

      class
0      dog_bark
1  children_playing
2  children_playing
3  children_playing
4  children_playing
5  children_playing
6  children_playing
7  children_playing
8  children_playing
9      car_horn
```

```
[10]: ### Check whether the dataset is imbalanced
      metadata['class'].value_counts()
```

```
[10]: class
      dog_bark          1000
      children_playing    1000
      air_conditioner     1000
      street_music        1000
      jackhammer          1000
      engine_idling       1000
      drilling            1000
      siren                929
      car_horn             429
      gun_shot             374
      Name: count, dtype: int64
```

```
[11]: dat1, sampling_rate1 = librosa.load('./audio/fold5/100032-3-0-0.wav')
      dat2, sampling_rate2 = librosa.load('./audio/fold5/100263-2-0-117.wav')
```

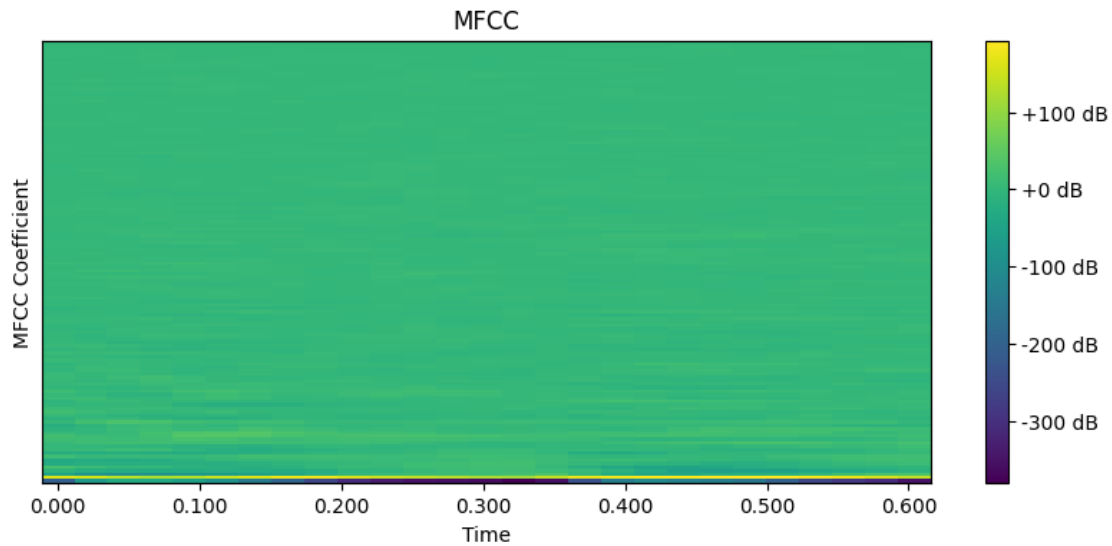
```
[27]: import librosa
import librosa.display
import matplotlib.pyplot as plt

def plot_mfcc(audio_path):
    # Load the audio file
    y, sr = librosa.load(audio_path)

    # Extract MFCC features
    mfccs = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=128)

    # Plot MFCCs
    plt.figure(figsize=(10, 4))
    librosa.display.specshow(mfccs, x_axis='time', cmap='viridis')
    plt.colorbar(format='%+2.0f dB')
    plt.title('MFCC')
    plt.xlabel('Time')
    plt.ylabel('MFCC Coefficient')
    plt.show()

# Example usage
audio_file_path = './audio/fold6/4910-3-0-0.wav'
plot_mfcc(audio_file_path)
```



```
[14]: import librosa#random sample
filename='./audio/fold6/4910-3-0-0.wav'
audio_data,sample_rate=librosa.load(filename)
mfccs=librosa.feature.mfcc(y=audio_data,sr=sample_rate, n_mfcc=128)
```

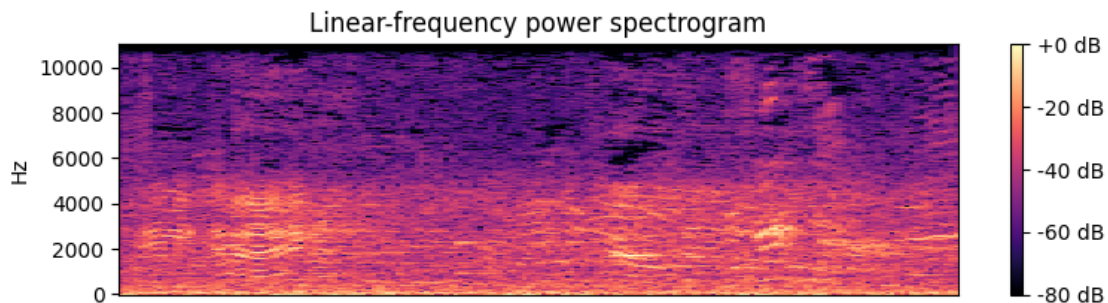


```
print(mfccs.shape)
print(mfccs)
```

```
(128, 27)
[[-2.0414842e+02 -7.3352737e+01 -3.1521664e+01 ... -2.8262634e+02
  -3.0813928e+02 -3.4986868e+02]
 [ 1.6085672e+02  1.5085524e+02  1.3205711e+02 ...  1.7034235e+02
  1.5520447e+02  1.3646313e+02]
 [-4.7794083e+01 -8.1008530e+01 -9.6134056e+01 ...  1.7801125e+01
  2.7814308e+01  3.0667643e+01]
 ...
 [ 1.8539858e-01  1.7941580e+00  5.5091381e-01 ...  7.5651646e-01
  2.4906158e-02  5.3799915e-01]
 [-7.3656654e-01 -1.1901817e+00 -9.0071106e-01 ...  4.8907185e-01
  -1.5518656e+00 -1.2077427e+00]
 [ 8.4392548e-01  3.3357239e-01 -2.3057556e-01 ...  4.3945312e-01
  1.3983688e+00  2.5930862e+00]]
```

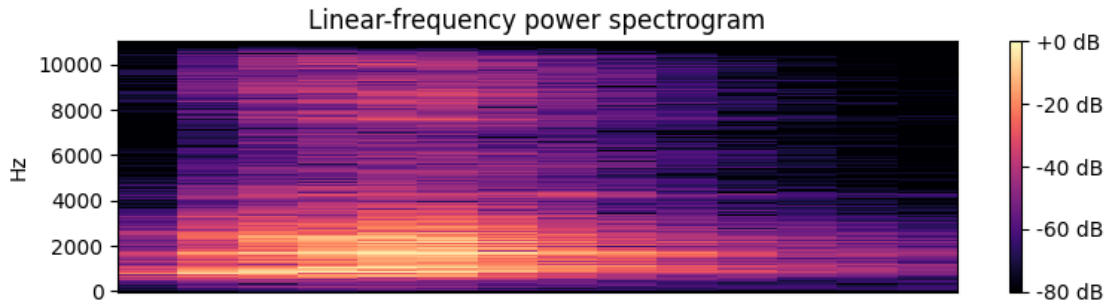
```
[13]: plt.figure(figsize=(20, 10))
D = librosa.amplitude_to_db(np.abs(librosa.stft(dat2)), ref=np.max)
plt.subplot(4, 2, 1)
librosa.display.specshow(D, y_axis='linear')
plt.colorbar(format='%+2.0f dB')
plt.title('Linear-frequency power spectrogram')
```

```
[13]: Text(0.5, 1.0, 'Linear-frequency power spectrogram')
```



```
[12]: plt.figure(figsize=(20, 10))
D = librosa.amplitude_to_db(np.abs(librosa.stft(dat1)), ref=np.max)
plt.subplot(4, 2, 1)
librosa.display.specshow(D, y_axis='linear')
plt.colorbar(format='%+2.0f dB')
plt.title('Linear-frequency power spectrogram')
```

```
[12]: Text(0.5, 1.0, 'Linear-frequency power spectrogram')
```



```
[10]: ### Let's read a sample audio using librosa
import librosa
librosa_audio_data,librosa_sample_rate=librosa.load(filename)
print(librosa_audio_data)
```

```
[-1.0477379e-08  9.3132257e-10  9.3132257e-10 ...  5.2534759e-02
-5.0471686e-03 -2.0513367e-02]
```

```
[20]: import numpy as np
from tqdm import tqdm
import os

def features_extractor(file):
    audio, sample_rate = librosa.load(file_name)
    mfccs_features = librosa.feature.mfcc(y=audio, sr=sample_rate, n_mfcc=128)
    mfccs_scaled_features = np.mean(mfccs_features.T,axis=0)
    return mfccs_scaled_features

# Now we iterate through every audio file and extract features
# using Mel-Frequency Cepstral Coefficients

audio_dataset_path='audio/'
extracted_features=[]
metadata = pd.read_csv('./audio/UrbanSound8K.csv')
for index_num,row in tqdm(metadata.iterrows()):
    file_name = os.path.join(os.path.
↪abspath(audio_dataset_path),'fold'+str(row["fold"])+ '/'
↪',str(row["slice_file_name"])))

    final_class_labels=row["class"]

    data=features_extractor(file_name)
    extracted_features.append([data,final_class_labels])

# converting extracted_features to Pandas dataframe
```

```

extracted_features_df=pd.
↳DataFrame(extracted_features,columns=['feature','class'])
extracted_features_df.head()

```

```

3554it [01:21, 42.06it/s]/home/ubuntu/MyProject/env/lib/python3.10/site-
packages/librosa/core/spectrum.py:266: UserWarning: n_fft=2048 is too large for
input signal of length=1323
    warnings.warn(
8324it [03:16, 56.14it/s]/home/ubuntu/MyProject/env/lib/python3.10/site-
packages/librosa/core/spectrum.py:266: UserWarning: n_fft=2048 is too large for
input signal of length=1103
    warnings.warn(
/home/ubuntu/MyProject/env/lib/python3.10/site-
packages/librosa/core/spectrum.py:266: UserWarning: n_fft=2048 is too large for
input signal of length=1523
    warnings.warn(
8732it [03:25, 42.39it/s]

```

```

[20]:

```

	feature	class
0	[-211.93698, 62.58121, -122.813156, -60.745293...	dog_bark
1	[-417.0052, 99.336624, -42.995583, 51.073326, ...	children_playing
2	[-452.39316, 112.36253, -37.57807, 43.195866, ...	children_playing
3	[-406.47922, 91.1966, -25.043556, 42.78452, 11...	children_playing
4	[-439.63873, 103.86223, -42.658783, 50.690285,...	children_playing

```

[29]: ### Split the dataset into independent and dependent dataset
X=np.array(extracted_features_df['feature'].tolist())
y=np.array(extracted_features_df['class'].tolist())
print(X.shape)

```

```

(8732, 128)

```

```

[31]: # Label Encoder
from tensorflow.keras.utils import to_categorical
from sklearn.preprocessing import LabelEncoder
labelencoder=LabelEncoder()
y=to_categorical(labelencoder.fit_transform(y))
print(y)

```

```

2025-05-02 18:29:13.214846: I external/local_xla/xla/tsl/cuda/cudart_stub.cc:32]
Could not find cuda drivers on your machine, GPU will not be used.
2025-05-02 18:29:13.220120: I external/local_xla/xla/tsl/cuda/cudart_stub.cc:32]
Could not find cuda drivers on your machine, GPU will not be used.
2025-05-02 18:29:13.232541: E
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register
cuFFT factory: Attempting to register factory for plugin cuFFT when one has
already been registered
WARNING: All log messages before absl::InitializeLog() is called are written to

```

STDERR

E0000 00:00:1746210553.251400 3634 cuda_dnn.cc:8579] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been registered

E0000 00:00:1746210553.256790 3634 cuda_blas.cc:1407] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already been registered

W0000 00:00:1746210553.271560 3634 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.

W0000 00:00:1746210553.271583 3634 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.

W0000 00:00:1746210553.271585 3634 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.

W0000 00:00:1746210553.271587 3634 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.

2025-05-02 18:29:13.276517: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.

To enable the following instructions: AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.

```
[[0. 0. 0. ... 0. 0. 0.]
 [0. 0. 1. ... 0. 0. 0.]
 [0. 0. 1. ... 0. 0. 0.]
 ...
 [0. 1. 0. ... 0. 0. 0.]
 [0. 1. 0. ... 0. 0. 0.]
 [0. 1. 0. ... 0. 0. 0.]]
```

```
[32]: ### Train Test Split
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=0)
print(X_train)
print(X_train.shape) #(6985, 128)
print(X_test.shape) #(1747, 128)
print(y_train.shape) #(6985, 10)
print(y_test.shape) #(1747, 10)
```

```
[[ -1.24484680e+02  1.03169846e+02 -1.33165455e+01 ... -9.90453303e-01
   -9.43154037e-01 -1.02599645e+00]
 [ -5.13921928e+00  7.90478210e+01  4.16248989e+00 ... -3.82960051e-01
   -8.50546658e-02 -2.49319136e-01]
 [ -4.03939552e+01 -1.31092958e+01 -7.20987606e+00 ... -9.23266172e-01
   -1.03576042e-01  6.51953161e-01]]
```

```
...
[-4.26655701e+02  9.15513382e+01  4.19170284e+00 ...  4.15096909e-01
 2.18663603e-01  2.13547856e-01]
[-1.37251968e+02  1.24268539e+02 -2.15997734e+01 ...  8.00883248e-02
 1.33612096e-01  2.86800683e-01]
[-4.18389374e+02  2.06862305e+02  7.23461866e+00 ... -1.40962983e-02
 -6.32517561e-02 -1.06617615e-01]]
(6985, 128)
(1747, 128)
(6985, 10)
(1747, 10)
```

```
[35]: import pandas as pd
import numpy as np

pd.plotting.register_matplotlib_converters()
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from sklearn.model_selection import GridSearchCV

from sklearn.preprocessing import MinMaxScaler
```

```
[36]: # Libraries for Classification and building Models

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, Flatten, Dense, MaxPool2D, Dropout
from tensorflow.keras.utils import to_categorical

from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
```

```
[40]: # Project Specific Libraries

import os
import librosa
import librosa.display
import glob
import skimage
```

```
[41]: X_train = X_train.reshape(6985, 16, 8, 1)
X_test = X_test.reshape(1747, 16, 8, 1)
```

```
[42]: input_dim = (16, 8, 1)
```

```
[43]: model = Sequential()
```

```
[44]: model.add(Conv2D(64, (3, 3), padding = "same", activation = "tanh", input_shape=(
    ↳ input_dim))
model.add(MaxPool2D(pool_size=(2, 2)))
model.add(Conv2D(128, (3, 3), padding = "same", activation = "tanh"))
model.add(MaxPool2D(pool_size=(2, 2)))
model.add(Dropout(0.1))
model.add(Flatten())
model.add(Dense(1024, activation = "tanh"))
model.add(Dense(10, activation = "softmax"))
```

/home/ubuntu/MyProject/env/lib/python3.10/site-packages/keras/src/layers/convolutional/base_conv.py:107: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
2025-05-02 18:41:03.332527: E
external/local_xla/xla/stream_executor/cuda/cuda_platform.cc:51] failed call to
cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
```

```
[45]: model.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics = [
    ↳ ['accuracy']])
```

```
[47]: model.fit(X_train, y_train, epochs = 90, batch_size = 50, validation_data = (
    ↳ (X_test, y_test))
```

```
Epoch 1/90
140/140          5s 28ms/step -
accuracy: 0.4567 - loss: 1.6437 - val_accuracy: 0.7670 - val_loss: 0.7238
Epoch 2/90
140/140          4s 27ms/step -
accuracy: 0.7934 - loss: 0.5976 - val_accuracy: 0.8105 - val_loss: 0.5789
Epoch 3/90
140/140          4s 26ms/step -
accuracy: 0.8668 - loss: 0.3924 - val_accuracy: 0.8392 - val_loss: 0.4600
Epoch 4/90
140/140          4s 26ms/step -
accuracy: 0.8964 - loss: 0.2962 - val_accuracy: 0.8558 - val_loss: 0.4545
Epoch 5/90
140/140          4s 26ms/step -
accuracy: 0.9152 - loss: 0.2570 - val_accuracy: 0.8666 - val_loss: 0.4709
Epoch 6/90
140/140          4s 25ms/step -
accuracy: 0.9254 - loss: 0.2192 - val_accuracy: 0.8500 - val_loss: 0.4977
Epoch 7/90
```

140/140 4s 26ms/step -
 accuracy: 0.9423 - loss: 0.1647 - val_accuracy: 0.8781 - val_loss: 0.4947
 Epoch 8/90
 140/140 4s 27ms/step -
 accuracy: 0.9557 - loss: 0.1267 - val_accuracy: 0.8975 - val_loss: 0.4330
 Epoch 9/90
 140/140 4s 27ms/step -
 accuracy: 0.9621 - loss: 0.1085 - val_accuracy: 0.8890 - val_loss: 0.4716
 Epoch 10/90
 140/140 4s 27ms/step -
 accuracy: 0.9602 - loss: 0.1160 - val_accuracy: 0.8798 - val_loss: 0.5049
 Epoch 11/90
 140/140 4s 31ms/step -
 accuracy: 0.9599 - loss: 0.1235 - val_accuracy: 0.8661 - val_loss: 0.5669
 Epoch 12/90
 140/140 3s 25ms/step -
 accuracy: 0.9671 - loss: 0.0987 - val_accuracy: 0.8872 - val_loss: 0.4716
 Epoch 13/90
 140/140 4s 31ms/step -
 accuracy: 0.9602 - loss: 0.1134 - val_accuracy: 0.8941 - val_loss: 0.4914
 Epoch 14/90
 140/140 4s 31ms/step -
 accuracy: 0.9746 - loss: 0.0723 - val_accuracy: 0.9004 - val_loss: 0.4565
 Epoch 15/90
 140/140 4s 31ms/step -
 accuracy: 0.9802 - loss: 0.0529 - val_accuracy: 0.8861 - val_loss: 0.5333
 Epoch 16/90
 140/140 4s 29ms/step -
 accuracy: 0.9762 - loss: 0.0665 - val_accuracy: 0.8912 - val_loss: 0.5528
 Epoch 17/90
 140/140 4s 29ms/step -
 accuracy: 0.9719 - loss: 0.0792 - val_accuracy: 0.8792 - val_loss: 0.6013
 Epoch 18/90
 140/140 4s 27ms/step -
 accuracy: 0.9770 - loss: 0.0717 - val_accuracy: 0.8958 - val_loss: 0.5302
 Epoch 19/90
 140/140 4s 30ms/step -
 accuracy: 0.9866 - loss: 0.0401 - val_accuracy: 0.8701 - val_loss: 0.6473
 Epoch 20/90
 140/140 4s 29ms/step -
 accuracy: 0.9748 - loss: 0.0726 - val_accuracy: 0.9078 - val_loss: 0.5177
 Epoch 21/90
 140/140 4s 29ms/step -
 accuracy: 0.9823 - loss: 0.0503 - val_accuracy: 0.9015 - val_loss: 0.5494
 Epoch 22/90
 140/140 4s 30ms/step -
 accuracy: 0.9735 - loss: 0.0846 - val_accuracy: 0.8930 - val_loss: 0.5823
 Epoch 23/90

140/140 4s 31ms/step -
 accuracy: 0.9887 - loss: 0.0324 - val_accuracy: 0.8935 - val_loss: 0.5472
 Epoch 24/90
 140/140 4s 29ms/step -
 accuracy: 0.9912 - loss: 0.0273 - val_accuracy: 0.9107 - val_loss: 0.5363
 Epoch 25/90
 140/140 4s 30ms/step -
 accuracy: 0.9929 - loss: 0.0234 - val_accuracy: 0.9118 - val_loss: 0.5117
 Epoch 26/90
 140/140 4s 29ms/step -
 accuracy: 0.9941 - loss: 0.0167 - val_accuracy: 0.9038 - val_loss: 0.5614
 Epoch 27/90
 140/140 4s 30ms/step -
 accuracy: 0.9813 - loss: 0.0583 - val_accuracy: 0.8752 - val_loss: 0.6987
 Epoch 28/90
 140/140 4s 29ms/step -
 accuracy: 0.9753 - loss: 0.0866 - val_accuracy: 0.9078 - val_loss: 0.5748
 Epoch 29/90
 140/140 4s 30ms/step -
 accuracy: 0.9897 - loss: 0.0347 - val_accuracy: 0.8844 - val_loss: 0.7185
 Epoch 30/90
 140/140 4s 29ms/step -
 accuracy: 0.9889 - loss: 0.0343 - val_accuracy: 0.8952 - val_loss: 0.6309
 Epoch 31/90
 140/140 4s 30ms/step -
 accuracy: 0.9872 - loss: 0.0364 - val_accuracy: 0.8958 - val_loss: 0.6287
 Epoch 32/90
 140/140 4s 30ms/step -
 accuracy: 0.9914 - loss: 0.0236 - val_accuracy: 0.9073 - val_loss: 0.5561
 Epoch 33/90
 140/140 4s 30ms/step -
 accuracy: 0.9884 - loss: 0.0289 - val_accuracy: 0.8827 - val_loss: 0.6923
 Epoch 34/90
 140/140 4s 31ms/step -
 accuracy: 0.9828 - loss: 0.0480 - val_accuracy: 0.9078 - val_loss: 0.5653
 Epoch 35/90
 140/140 4s 31ms/step -
 accuracy: 0.9845 - loss: 0.0408 - val_accuracy: 0.8970 - val_loss: 0.6623
 Epoch 36/90
 140/140 4s 30ms/step -
 accuracy: 0.9856 - loss: 0.0370 - val_accuracy: 0.8998 - val_loss: 0.6179
 Epoch 37/90
 140/140 5s 34ms/step -
 accuracy: 0.9857 - loss: 0.0393 - val_accuracy: 0.8998 - val_loss: 0.6541
 Epoch 38/90
 140/140 5s 34ms/step -
 accuracy: 0.9899 - loss: 0.0241 - val_accuracy: 0.9010 - val_loss: 0.6430
 Epoch 39/90

140/140 5s 34ms/step -
 accuracy: 0.9888 - loss: 0.0294 - val_accuracy: 0.9078 - val_loss: 0.6017
 Epoch 40/90
 140/140 5s 34ms/step -
 accuracy: 0.9852 - loss: 0.0430 - val_accuracy: 0.9038 - val_loss: 0.6359
 Epoch 41/90
 140/140 5s 34ms/step -
 accuracy: 0.9914 - loss: 0.0215 - val_accuracy: 0.8815 - val_loss: 0.7264
 Epoch 42/90
 140/140 4s 31ms/step -
 accuracy: 0.9928 - loss: 0.0228 - val_accuracy: 0.9067 - val_loss: 0.5969
 Epoch 43/90
 140/140 4s 30ms/step -
 accuracy: 0.9904 - loss: 0.0294 - val_accuracy: 0.8947 - val_loss: 0.6256
 Epoch 44/90
 140/140 4s 30ms/step -
 accuracy: 0.9882 - loss: 0.0331 - val_accuracy: 0.8993 - val_loss: 0.6349
 Epoch 45/90
 140/140 4s 30ms/step -
 accuracy: 0.9741 - loss: 0.0847 - val_accuracy: 0.9090 - val_loss: 0.5522
 Epoch 46/90
 140/140 4s 30ms/step -
 accuracy: 0.9888 - loss: 0.0289 - val_accuracy: 0.9021 - val_loss: 0.5985
 Epoch 47/90
 140/140 4s 31ms/step -
 accuracy: 0.9896 - loss: 0.0266 - val_accuracy: 0.9136 - val_loss: 0.5850
 Epoch 48/90
 140/140 5s 34ms/step -
 accuracy: 0.9927 - loss: 0.0193 - val_accuracy: 0.9096 - val_loss: 0.6083
 Epoch 49/90
 140/140 5s 34ms/step -
 accuracy: 0.9931 - loss: 0.0238 - val_accuracy: 0.9113 - val_loss: 0.5780
 Epoch 50/90
 140/140 4s 30ms/step -
 accuracy: 0.9956 - loss: 0.0151 - val_accuracy: 0.9118 - val_loss: 0.5997
 Epoch 51/90
 140/140 4s 31ms/step -
 accuracy: 0.9971 - loss: 0.0083 - val_accuracy: 0.9136 - val_loss: 0.5879
 Epoch 52/90
 140/140 4s 31ms/step -
 accuracy: 0.9969 - loss: 0.0104 - val_accuracy: 0.9227 - val_loss: 0.5469
 Epoch 53/90
 140/140 4s 31ms/step -
 accuracy: 0.9943 - loss: 0.0252 - val_accuracy: 0.8890 - val_loss: 0.6708
 Epoch 54/90
 140/140 4s 30ms/step -
 accuracy: 0.9831 - loss: 0.0503 - val_accuracy: 0.8867 - val_loss: 0.7081
 Epoch 55/90

140/140 4s 30ms/step -
 accuracy: 0.9810 - loss: 0.0615 - val_accuracy: 0.8970 - val_loss: 0.7757
 Epoch 56/90
 140/140 4s 31ms/step -
 accuracy: 0.9811 - loss: 0.0610 - val_accuracy: 0.9056 - val_loss: 0.6334
 Epoch 57/90
 140/140 4s 31ms/step -
 accuracy: 0.9937 - loss: 0.0159 - val_accuracy: 0.9067 - val_loss: 0.5909
 Epoch 58/90
 140/140 5s 32ms/step -
 accuracy: 0.9947 - loss: 0.0167 - val_accuracy: 0.9101 - val_loss: 0.5571
 Epoch 59/90
 140/140 4s 30ms/step -
 accuracy: 0.9968 - loss: 0.0100 - val_accuracy: 0.9044 - val_loss: 0.6297
 Epoch 60/90
 140/140 4s 27ms/step -
 accuracy: 0.9954 - loss: 0.0134 - val_accuracy: 0.9015 - val_loss: 0.5917
 Epoch 61/90
 140/140 4s 27ms/step -
 accuracy: 0.9943 - loss: 0.0165 - val_accuracy: 0.9027 - val_loss: 0.6478
 Epoch 62/90
 140/140 4s 30ms/step -
 accuracy: 0.9912 - loss: 0.0331 - val_accuracy: 0.8918 - val_loss: 0.6506
 Epoch 63/90
 140/140 4s 26ms/step -
 accuracy: 0.9879 - loss: 0.0443 - val_accuracy: 0.8998 - val_loss: 0.6783
 Epoch 64/90
 140/140 4s 26ms/step -
 accuracy: 0.9919 - loss: 0.0218 - val_accuracy: 0.8907 - val_loss: 0.6951
 Epoch 65/90
 140/140 4s 27ms/step -
 accuracy: 0.9912 - loss: 0.0345 - val_accuracy: 0.9004 - val_loss: 0.6674
 Epoch 66/90
 140/140 4s 26ms/step -
 accuracy: 0.9950 - loss: 0.0147 - val_accuracy: 0.9050 - val_loss: 0.5606
 Epoch 67/90
 140/140 4s 27ms/step -
 accuracy: 0.9949 - loss: 0.0154 - val_accuracy: 0.9130 - val_loss: 0.6131
 Epoch 68/90
 140/140 4s 28ms/step -
 accuracy: 0.9956 - loss: 0.0111 - val_accuracy: 0.9004 - val_loss: 0.6841
 Epoch 69/90
 140/140 4s 28ms/step -
 accuracy: 0.9958 - loss: 0.0163 - val_accuracy: 0.9096 - val_loss: 0.5979
 Epoch 70/90
 140/140 4s 28ms/step -
 accuracy: 0.9942 - loss: 0.0161 - val_accuracy: 0.8993 - val_loss: 0.6597
 Epoch 71/90

140/140 4s 28ms/step -
 accuracy: 0.9877 - loss: 0.0364 - val_accuracy: 0.8895 - val_loss: 0.7454
 Epoch 72/90
 140/140 4s 28ms/step -
 accuracy: 0.9866 - loss: 0.0444 - val_accuracy: 0.9044 - val_loss: 0.6357
 Epoch 73/90
 140/140 4s 28ms/step -
 accuracy: 0.9931 - loss: 0.0203 - val_accuracy: 0.8998 - val_loss: 0.6793
 Epoch 74/90
 140/140 4s 28ms/step -
 accuracy: 0.9894 - loss: 0.0284 - val_accuracy: 0.9004 - val_loss: 0.6450
 Epoch 75/90
 140/140 5s 32ms/step -
 accuracy: 0.9952 - loss: 0.0179 - val_accuracy: 0.8941 - val_loss: 0.7158
 Epoch 76/90
 140/140 5s 34ms/step -
 accuracy: 0.9926 - loss: 0.0243 - val_accuracy: 0.9096 - val_loss: 0.6696
 Epoch 77/90
 140/140 4s 32ms/step -
 accuracy: 0.9931 - loss: 0.0180 - val_accuracy: 0.9050 - val_loss: 0.6257
 Epoch 78/90
 140/140 4s 31ms/step -
 accuracy: 0.9964 - loss: 0.0131 - val_accuracy: 0.9078 - val_loss: 0.6835
 Epoch 79/90
 140/140 4s 31ms/step -
 accuracy: 0.9970 - loss: 0.0116 - val_accuracy: 0.9096 - val_loss: 0.6225
 Epoch 80/90
 140/140 4s 31ms/step -
 accuracy: 0.9949 - loss: 0.0115 - val_accuracy: 0.8987 - val_loss: 0.7500
 Epoch 81/90
 140/140 4s 31ms/step -
 accuracy: 0.9936 - loss: 0.0200 - val_accuracy: 0.8981 - val_loss: 0.7455
 Epoch 82/90
 140/140 4s 32ms/step -
 accuracy: 0.9932 - loss: 0.0211 - val_accuracy: 0.8998 - val_loss: 0.6838
 Epoch 83/90
 140/140 4s 31ms/step -
 accuracy: 0.9966 - loss: 0.0094 - val_accuracy: 0.9050 - val_loss: 0.6892
 Epoch 84/90
 140/140 4s 31ms/step -
 accuracy: 0.9906 - loss: 0.0322 - val_accuracy: 0.9010 - val_loss: 0.6796
 Epoch 85/90
 140/140 4s 31ms/step -
 accuracy: 0.9939 - loss: 0.0180 - val_accuracy: 0.8998 - val_loss: 0.7826
 Epoch 86/90
 140/140 4s 31ms/step -
 accuracy: 0.9945 - loss: 0.0196 - val_accuracy: 0.8981 - val_loss: 0.8058
 Epoch 87/90

```

140/140          4s 31ms/step -
accuracy: 0.9944 - loss: 0.0177 - val_accuracy: 0.9033 - val_loss: 0.7487
Epoch 88/90
140/140          4s 31ms/step -
accuracy: 0.9917 - loss: 0.0205 - val_accuracy: 0.8958 - val_loss: 0.7274
Epoch 89/90
140/140          4s 31ms/step -
accuracy: 0.9905 - loss: 0.0274 - val_accuracy: 0.8964 - val_loss: 0.7613
Epoch 90/90
140/140          4s 31ms/step -
accuracy: 0.9930 - loss: 0.0202 - val_accuracy: 0.8935 - val_loss: 0.8262

```

[47]: <keras.src.callbacks.history.History at 0x7b5364368bb0>

[48]: `model.summary()`

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 16, 8, 64)	640
max_pooling2d (MaxPooling2D)	(None, 8, 4, 64)	0
conv2d_1 (Conv2D)	(None, 8, 4, 128)	73,856
max_pooling2d_1 (MaxPooling2D)	(None, 4, 2, 128)	0
dropout (Dropout)	(None, 4, 2, 128)	0
flatten (Flatten)	(None, 1024)	0
dense (Dense)	(None, 1024)	1,049,600
dense_1 (Dense)	(None, 10)	10,250

Total params: 3,403,040 (12.98 MB)

Trainable params: 1,134,346 (4.33 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 2,268,694 (8.65 MB)

```
[49]: predictions = model.predict(X_test)
      score = model.evaluate(X_test, y_test)
      print(score)
```

```
55/55          0s 6ms/step
55/55          0s 5ms/step -
accuracy: 0.8841 - loss: 0.9057
[0.8261630535125732, 0.8935317397117615]
```

```
[50]: preds = np.argmax(predictions, axis = 1)
```

```
[52]: df_result = pd.DataFrame(preds)
      df_result.to_csv("UrbanSound8kResults.csv")
```

```
[ ]:
```